

Gaze-Pinch Menu: Performing Multiple Interactions Concurrently in Mixed Reality

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ABSTRACT

Performing an interaction using gaze and pinch has been certified as an efficient interactive method in Mixed Reality, for such techniques can provide users concise and natural experiences. However, executing a task with individual interactions gradually is inefficient in some application scenarios. In this paper, we propose the Hand-Pinch Menu, which core concept is to reduce unnecessary operations by combining several interactions. Users can continuously perform multiple interactions on a selected object concurrently without changing gestures by using this technique. The user study results show that our Gaze-Pinch Menu can improve operational efficiency effectively.

Index Terms: Human-centered computing—Human computer interaction (HCI); Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality

1 INTRODUCTION

Interacting with virtual objects using gaze and bare-hand gestures is a concise and efficient paradigm for mixed reality (MR) and augmented reality (AR). The core of gaze-gesture interaction is to employ the gaze as a cursor and use specific gestures to device controller buttons to confirm the selection or manipulate virtual objects [3]. The graphical menu [1, 4] is the most common solution for transformation between independent interactions. However, considering a specific gesture can be used as the metaphor of different interactions depending on the context, object properties, and user intention, performing multiple interactions concurrently with the same gesture can make more benefits in some cases.

In this paper, we propose the Gaze-Pinch Menu (GPMenu) as a tool to handle the selection and implementation of interactions, and the GPMenu can be further subdivided into the Single-hand Gaze-Pinch Menu (SGPMenu) and the Bimanual Gaze-Pinch Menu (BGPMenu) (Figure 1). GPMenu is an object-oriented menu. The application range of GPMenu is restricted to the selected gaze target, and the selection of interaction modes is decided by holding a pinch gesture and moving the corresponding hand.

We conduct a comparative study of a traditional gaze-pinch-based menu [5] and our approach with two types of tasks: scaling and coloring. Results show that both SGPMenu and BGPMenu can improve efficiency, and BGPMenu is more acceptable for users.

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Figure 1: Manipulating objects in mixed reality with Gaze-Pinch Menu.

2 RELATED WORK

Eye-tracking is considered to be a convenient and natural interaction interface. However, the Midas Touch problem [2] has restricted the application scenarios of eye-gaze interaction in the human-computer interaction (HCI) domain seriously. The graphical menu [1, 4] is a common used interaction designated method in MR.

Most graphical menus are process-oriented [1], whatever the menu is controlled by bare-hand gestures or hardware peripherals. The standard operation processes can be summarized as the selection stage and the execution stage. These menus are quite effective, but are not suitable for uninterrupted tasks, since a complex task usually contains multiple interactions.

Mardanbegi et al. [4] offered the other design concept of the MR menu. They proposed an object-oriented menu for VR interaction, called EyeSeeThrough. That technique achieved unified interaction mode selection and application by utilizing the ability to align the virtual object and tool relative to each other using an eye-gaze ray. Our GPMenu is based on the same conception, and supporting mode-switching [6] without changing the target.

3 GAZE-PINCH MENU

Gaze-Pinch Menu is proposed to select interactions during the execution of a complex task in MR environments, and it enabled users performing multiple interactions concurrently to improve efficiency. In terms of mode-switching methods, the proposed menu can be further subdivided into two interaction techniques: SGPMenu and BGPMenu. To a selected target object, the operation process in completing a complex task can be summarized as five steps (Figure 2).

Confirm Interaction Target. The first step is to determine the target object. Our GPMenu employs gaze to achieve the goal of selecting a target. The gaze is manifested as a translucent sphere on the virtual object's surface that is looking at by a user. Users can then confirm the selection by making a pinch gesture with thumb and index.

Switch to Menu Interface. Users need to select interactions they want by switching to the menu interface. Specifically, we convert the pinching hand into a coordinate that defines users' right-hand direction as the x-axis, the upward direction as the y-axis, and the forward direction as the z-axis. In SGPMenu, the model transforms

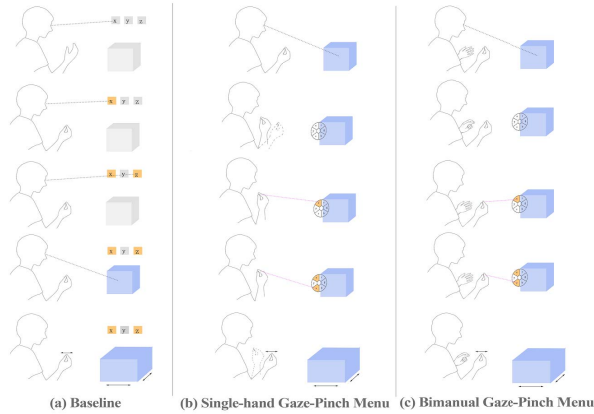


Figure 2: Scaling a virtual cube with Baseline (a), Single-Hand Gaze-Pinch Menu (b), and Bimanual Gaze-Pinch Menu (c). All of these menus support multiple selections, and the pinching hand's movement on the x-axis controlling the value of scale. SGPMenu achieving mode-switching by moving backward on the z-axis, while BGPMenu does the same by pinching the other hand.

from selection state to menu interface will be activated when the movement distance of the pinching hand Z_{Move} exceeds the threshold Z_{Menu} . While for BGPMenu, the mode switch trigger is set to the pinch gesture of the other hand.

Select Interactions. After transforming the menu interface to interaction selection, users can start choosing wanted interactions. We use a circular mark to remind the pinching hand's location on the x-axis and y-axis.

Switch to Operation Mode. The operation is similar to step 2. In SGPMenu, users should move their pinching hand until achieving the operation distance Z_{Operat} . For BGPMenu, the trigger of BGPMenu is still the pinch gesture of the other hand.

Operation and Repeat. Users can decide to complete the task by repeating step one to step four or "one take" freely.

4 USER STUDY

10 participants were recruited for this study (5 males and 5 females). Their average age was 23.1 (from 22 to 25, $SD = 0.89$), and all of them were students of our university. None of the participants had previous experience with head-worn virtual reality devices, but 3 participants had prior experience on mixed reality headsets (Hololens 2).

To examine the validity of SGPMenu and BGPMenu, we compared them with a widely used gaze-pinch-based interaction mode selecting method. The task was designed as scaling and coloring a given cube to a specific appearance. The target appearance will be shown to participants before the examination start. Considering the accuracy of color display, the color of objects was divided into 10 stages to avoid the issues of imperceptible change.

5 RESULTS

We used repeated measures ANOVAs for the analysis of our results. The completion time of tasks were regarded as measures.

There was a significant main effect on completion time for the baseline, SGPMenu, and BGPMenu ($F(2,59) = 3.19, p = 0.049$), as Figure 3 shows. The means and covariances ($M_{Baseline} = 65.37, SD_{Baseline} = 262.76, M_{SGPMenu} = 61.79, SD_{SGPMenu} = 449.13, M_{BGPMenu} = 51.96, SD_{BGPMenu} = 275.61$) shown that there were differences among participants, and BGPMenu had achieved the best performance. We also found a significant main effect between the single-interaction mode and the multiple-interaction mode

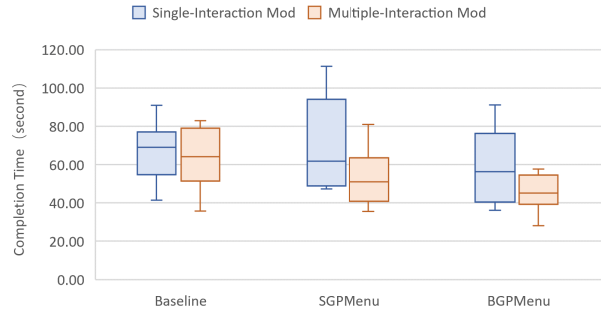


Figure 3: The Completion Time.

($F(1,59) = 6.73, p = 0.012$), and the means of them were $M_{Single} = 65.53$ ($SD_{Single} = 400.44$) and $M_{Multiple} = 53.89$ ($SD_{Multiple} = 242.96$) respectively. However, the results of ANOVAs ($F(2,59) = 0.699, p = 0.502$) shown that the mixture of these menus and the amount interaction was non-significant and irrelevant to user's operational efficiency.

6 DISCUSSION

The evaluation of users was closely related to proficiency. Some participants (3 of 10) have given quite different evaluations in two adjacent days. On the first day, these participants were required to complete the task and fill in the SUS form. The next day, we asked them to do the same experiment. The attitudes towards GPMMenu were changed from negative to neutral (1 of 3) or positive (2 of 3).

Users accept to execute multiple interactions simultaneously in some cases. The target appearance was obtained randomly in our experiment. One interesting fact was that participants usually manipulate the sub-interactions into two groups, the scale group and the color group.

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